Deciphering Equilibrium in Bayesian Rank-order Games with Costly Actions

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  - **Value interdependence**
  - **Multiple units**

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- However, equilibrium analysis in these settings can be challenging due to informational and allocative complexities:
  - **Uncertain number of participants**
  - **Value interdependence**
  - **Multiple units**
- The following features often present technical obstacles in establishing existence or characterizing equilibrium:
  - **Disconnected bidding supports** within a type
  - **Non-monotonicity** across types
  - **Atoms** in bidding distributions

What Do We Do in This Paper?

- We develop a novel approach to equilibrium analysis in a class of Bayesian rank-order games.

$$\text{Path (in a hypercube under some conditions)} \iff \text{BNE}$$

Model Setup

Primitives

- A fixed set of $n \geq 2$ players, $I = \{1, \ldots, n\}$, compete for $\ell \in \{1, \ldots, n\}$ prizes. All players participate.
- Player i privately observes $\theta_i \in \Theta_i$, where Θ_i is finite.
  - A common prior P on $\Theta = \prod_i \Theta_i$ gives every type positive marginal probability.
  - Types may be arbitrarily correlated. The joint prior need not have full support.
- The type profile $\boldsymbol{\theta}$ contains all payoff uncertainty. Payoffs may depend on opponents' types as well as one's own.
- After observing θ_i, players simultaneously choose actions $x_i \in \mathbb{R}_+$.

Model Setup: Bayesian Rank-order Games with Costly Actions

- Players rank in descending order of their actions. Rank 1 is the highest action.
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- A **fair random ordering** resolves every tie, including at zero action.
- The game allows interdependent values, multiple prizes, and non-separable utilities. Opponents' action levels matter only through rank.

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- For $x \in [0, \bar{x}]$, better ranks are weakly preferred:

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- The strict ranking-gain set

$$B_i(x, \boldsymbol{\theta}) := \left\{ r \in \{1, \ldots, n-1\} : u_i(x, r, \boldsymbol{\theta}) > u_i(x, r+1, \boldsymbol{\theta}) \right\}$$

is nonempty, and is the same across all $x \in [0, \bar{x}]$ and i.

Model Setup

Strategies and Equilibrium

- Index **player–type pairs** by $k \leftrightarrow (i, \theta_i)$, with $K = \sum_i |\Theta_i|$. Agent k uses a probability measure μ_k on $\mathbb{R}_+$ with CDF $F_k : \mathbb{R}_+ \mapsto [0, 1]$.

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- Denote the utility of choosing 0 and getting the last place by

$$\beta_k := \mathbb{E}[u_i(0, n, \boldsymbol{\theta}) \mid \theta_i] \geq 0.$$

Path-based Approach to Equilibrium Analysis

Probability Paths

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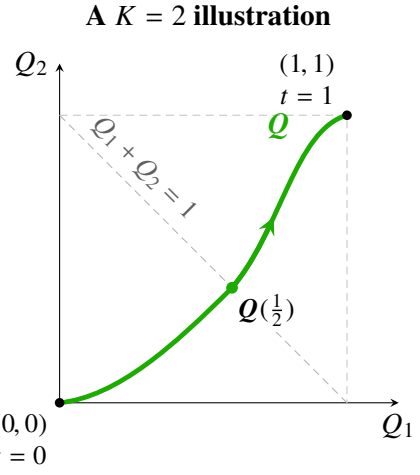
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- **The last requirement is simply a normalization: t, which parameterizes the curve, is taken to be the average of the coordinates of $\boldsymbol{Q}(t)$.**

Probability Paths

- At the moment, a probability path is just a geometric object that has nothing to do with the game or its equilibrium.
- We now show how probability paths can be used to analyze equilibria.



Equilibrium Representation

- Let $\nu_k = \mathrm{d}Q_k$ be the probability measure on $[0, 1]$ with CDF Q_k:

$$\nu_k((a, b]) = Q_k(b) - Q_k(a), \quad 0 \le a < b \le 1.$$

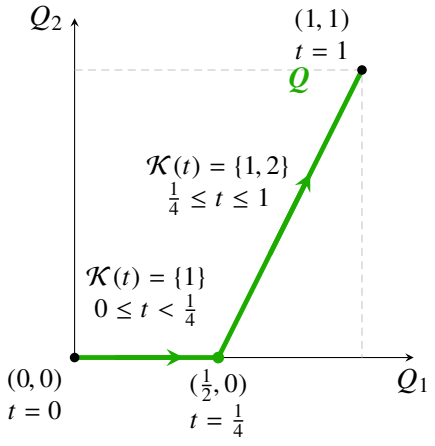
It records how much probability coordinate k accumulates along the path.

- The **active agents** at t are

$$\mathcal{K}(t) := \{k : t \in \operatorname{supp} \nu_k\}.$$

These are the coordinates that accumulate probability near t.

A $K = 2$ illustration



Equilibrium Representation

Recovered Actions and Incentive Compatibility

- Define the **tie-free payoff** $\widehat{U}_k(x; \boldsymbol{F})$ for $x \in \mathbb{R}_+$ and $\boldsymbol{F} \in [0,1]^K$ by giving opponent type (j, θ_j) probability F_{j,θ_j} of ranking below i, with no ties.

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- **Fix candidate payoffs $\boldsymbol{u} \geq \boldsymbol{\beta}$.** Define the action thresholds

$$\underline{x}_k(\boldsymbol{F}; u_k) := \inf\{x \geq 0 : \widehat{U}_k(x; \boldsymbol{F}) \leq u_k\}.$$

If an action satisfying $\widehat{U}_k(x; \boldsymbol{F}) = u_k$ exists, $\underline{x}_k(\boldsymbol{F}; u_k)$ is that action: holding $\boldsymbol{F}$ fixed, any lower (higher) action gives agent k a higher (lower) payoff than u_k.

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- **Definition:** We say that a path is ***incentive compatible*** if every active agent is indifferent: $\widehat{U}_k(\phi_{\boldsymbol{u}}(t);\boldsymbol{Q}(t)) = u_k$ for every $t \in [0,1]$ and $k \in \mathcal{K}(t)$.

Equilibrium Representation

Theorem 1 (Equilibrium and IC Paths)

Under Assumption 1:

(i) *Every BNE with payoff* $\boldsymbol{u}^*$ *and CDFs* F_k^* *corresponds to an IC path under* $\boldsymbol{u}^*$*, obtained from*

$$Q_k(t) = F_k^*(x) \quad \text{at} \quad t = \frac{1}{K}\sum\nolimits_{j=1}^{K} F_j^*(x), \qquad x \geq 0,$$

with $\boldsymbol{Q}(0) = \mathbf{0}$ *and linear interpolation across atoms. Moreover,* $\boldsymbol{u}^* \geq \boldsymbol{\beta}$ *and* $\min_k(u_k^* - \beta_k) = 0$.

(ii) *Every IC path with candidate payoffs* $\boldsymbol{u} \geq \boldsymbol{\beta}$ *determines a BNE of the Bayesian rank-order game, with payoff* $\boldsymbol{u}$*. Agent* k*'s mixed strategy has CDF*

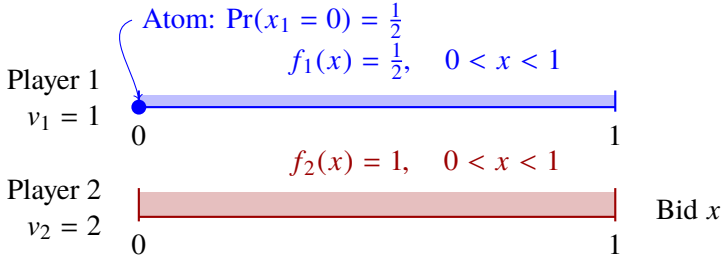
$$F_k(x) = \max\Big\{Q_k(t) : t \in [0,1],\ \phi_{\boldsymbol{u}}(t) \leq x\Big\}, \qquad x \geq 0.$$

Discussions of the Representation Theorem

- Theorem 1(i) is not surprising, we know that the strategy CDFs trace a path in the hypercube.

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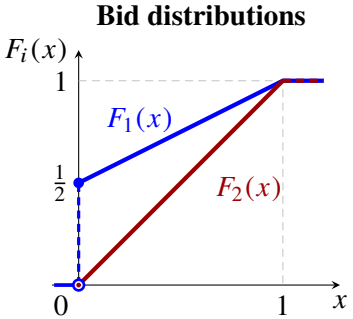
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  A simple example: Consider a complete-information all-pay auction with two players, where player 1 has value $v_1 = 1$ and player 2 has value $v_2 = 2$.



Both players have support $[0, 1]$; only player 1 has an atom at zero.

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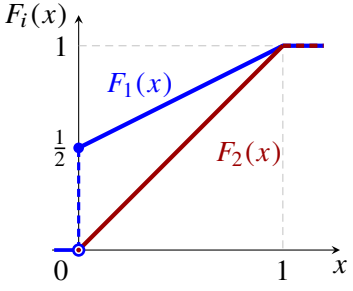


$F_1(x) = \frac{1}{2} + \frac{x}{2}, \quad F_2(x) = x, \quad 0 \le x \le 1.$

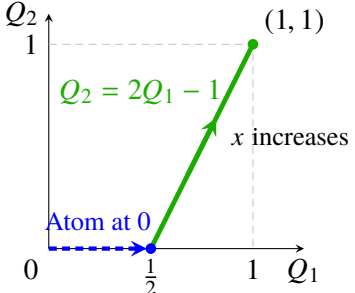
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Bid distributions



Path in distribution space



$F_1(x) = \frac{1}{2} + \frac{x}{2}$, $F_2(x) = x$, $0 \le x \le 1$. $(F_1(x), F_2(x)) = (\frac{1}{2} + \frac{x}{2}, x)$, $0 \le x \le 1$.

The dashed blue segment fills the jump caused by player 1's atom at $x = 0$.

Discussions of the Representation Theorem

- Theorem 1(i) is not surprising, we know that the strategy CDFs trace a path in the hypercube.
- Theorem 1(ii) is the crux: It tells us that **the path in the hypercube, together with a conjecture of u, contains sufficient information to check for equilibrium conditions.**

Discussions of the Representation Theorem

- Theorem 1(i) is not surprising, we know that the strategy CDFs trace a path in the hypercube.
- Theorem 1(ii) is the crux: It tells us that **the path in the hypercube, together with a conjecture of u, contains sufficient information to check for equilibrium conditions.**
- We proceed to make use of the path representation and establish equilibrium existence.

Path Existence

Existence of an Incentive-Compatible Path

Theorem 2 (Existence of an incentive-compatible path)

Under Assumption 1, there exist a payoff vector $\boldsymbol{u} \geq \boldsymbol{\beta}$ *satisfying* $\min_{1 \leq k \leq K}(u_k - \beta_k) = 0$ *and a probability path* $\boldsymbol{Q}$ *that is IC under* $\boldsymbol{u}$.

Intuition behind the Existence Theorem: Choosing Payoffs to Reach the Endpoint (i.e., Shooting)

- **For fixed $\boldsymbol{u}$, IC guides the path:** at $\boldsymbol{Q}(t)$, probability can accumulate only for agents whose tie-free payoffs equal u_k with action $\underline{x}(\boldsymbol{Q}(t);\boldsymbol{u})$:

$$\dot{\boldsymbol{Q}}(t) \in K \operatorname{conv}\left\{\boldsymbol{e}_k : \widehat{U}_k\Big(\underline{x}(\boldsymbol{Q}(t);\boldsymbol{u});\boldsymbol{Q}(t)\Big) = u_k\right\} \quad \text{a.e.}$$

  Here $\boldsymbol{e}_k$ is the kth coordinate vector.

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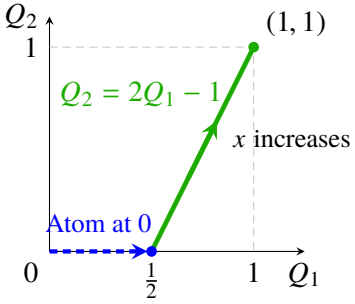
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- The endpoint varies with $\boldsymbol{u}$ so we can steer it toward $\mathbf{1}$ by adjusting $\boldsymbol{u}$ appropriately. A **topological degree** argument formalizes this intuition. ▸ details

Illustrative Example (continued)

- IC regulates how each position on the probability path is assigned to action x.
- For conjectured payoffs $\boldsymbol{u} \geq \mathbf{0}$, the recovered bid at $\boldsymbol{Q}(t)$ is $\underline{x}(t) := \max\left\{0,\ v_1 Q_2(t) - u_1,\ v_2 Q_1(t) - u_2\right\}$, which ensures that $v_1 Q_2(t) - \underline{x}(t) \leq u_1$ and $v_2 Q_1(t) - \underline{x}(t) \leq u_2$ for every $t \in [0,1]$.



- The path $\boldsymbol{Q}$ is **incentive compatible (IC)** under $\boldsymbol{u}$ if, for almost every $t \in [0,1]$,

$$Q_1'(t) > 0 \implies v_1 Q_2(t) - \underline{x}(t) = u_1,$$
$$Q_2'(t) > 0 \implies v_2 Q_1(t) - \underline{x}(t) = u_2.$$

Illustrative Example (continued)

- For a given candidate payoff vector (u_1, u_2), we can try to trace a path from $(0, 0)$ by following the IC conditions. If it hits $(1, 1)$, then we have found a BNE with payoffs (u_1, u_2). If it does not hit $(1, 1)$, then we try a different candidate payoff vector.

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- Specifically, at (Q_1, Q_2), the feasible directions are:

$$\begin{cases} \text{Increase only } Q_1, & \text{if } v_1 Q_2 - u_1 > v_2 Q_1 - u_2, \end{cases}$$

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Reason: In this case, we can set the action to $x = v_1 Q_2 - u_1$, so that $v_2 Q_1 - x < u_2$, consistent with player 2 being inactive.

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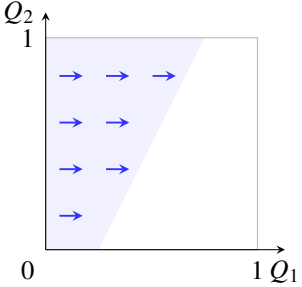
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Path Finding: Illustration

$v_1 = 1$, $v_2 = 2$; all directions are ok when $v_1 Q_2 - u_1 = v_2 Q_1 - u_2$.

Proposed payoffs

$\boldsymbol{u} = (0, 0.5)$



$v_1 Q_2 - u_1 > v_2 Q_1 - u_2$

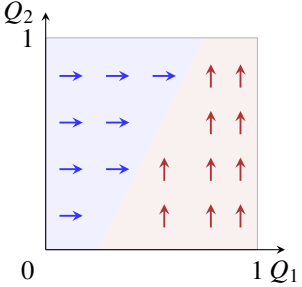
$\rightarrow$ Increase Q_1

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Proposed payoffs

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$v_1 Q_2 - u_1 < v_2 Q_1 - u_2$

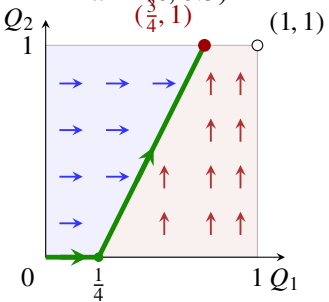
$\rightarrow$ Increase Q_1 $\uparrow$ Increase Q_2

Path Finding: Illustration

$v_1 = 1$, $v_2 = 2$; all directions are ok when $v_1 Q_2 - u_1 = v_2 Q_1 - u_2$.

Proposed payoffs

$\boldsymbol{u} = (0, 0.5)$



Cannot continue to $(1, 1)$:

Overshooting along Q_2.

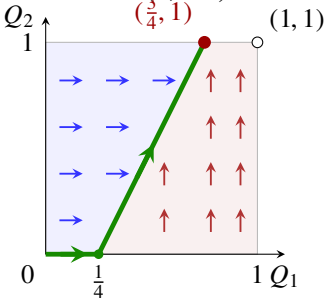
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Proposed payoffs

$\boldsymbol{u} = (0, 0.5)$

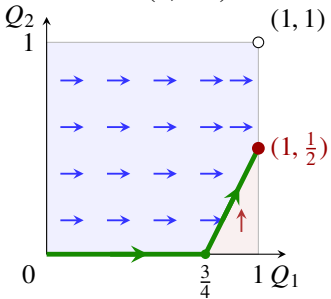


Cannot continue to $(1, 1)$:

Overshooting along Q_2.

Proposed payoffs

$\boldsymbol{u} = (0, 1.5)$



Cannot continue to $(1, 1)$:

Undershooting along Q_2.

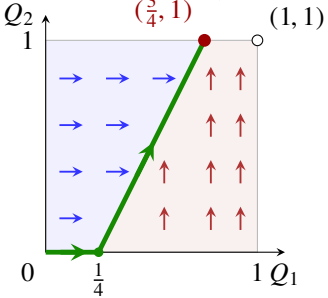
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Path Finding: Illustration

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Proposed payoffs

$\boldsymbol{u} = (0, 0.5)$

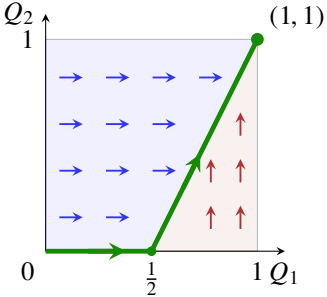


Cannot continue to $(1, 1)$:

Overshooting along Q_2.

Proposed payoffs

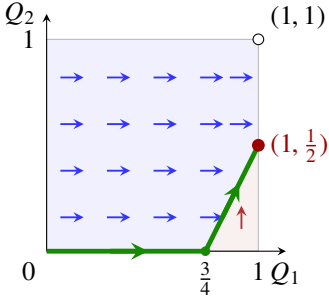
$\boldsymbol{u} = (0, 1)$



Reaches $(1, 1)$: an IC path.

Proposed payoffs

$\boldsymbol{u} = (0, 1.5)$



Cannot continue to $(1, 1)$:

Undershooting along Q_2.

$\rightarrow$ Increase Q_1 $\uparrow$ Increase Q_2

Equilibrium Existence

Corollary 1 (Equilibrium existence)

Under Assumption 1, the Bayesian rank-order game has a BNE. Its payoff vector satisfies $\boldsymbol{u} \geq \boldsymbol{\beta}$ *and* $\min_k (u_k - \beta_k) = 0$.

Application: All-Pay Auction

Assumption 2 (All-pay Auction Primitives)

There are n participating bidders and $\ell \in \{1, \ldots, n-1\}$ prizes. Values and costs may depend on the full type profile: $u_i(x, r, \boldsymbol{\theta}) = v_i^{(r)}(\boldsymbol{\theta}) - c_i(x, \boldsymbol{\theta})$.

- Bounded, weakly ordered values: $\overline{v} \geq v_i^{(1)} \geq \cdots \geq v_i^{(n)} \geq 0$, with $v_i^{(r)} = 0$ for $r > \ell$.
- $c_i(\cdot, \boldsymbol{\theta})$ is continuous and strictly increasing, with $c_i(0, \boldsymbol{\theta}) = 0$ and a common cutoff satisfying $c_i(\bar{x}, \boldsymbol{\theta}) > \overline{v}$.
- At each positive-probability profile, the sets

$$G_i(\boldsymbol{\theta}) := \{1 \leq r \leq \ell : v_i^{(r)}(\boldsymbol{\theta}) > v_i^{(r+1)}(\boldsymbol{\theta})\}$$

  are nonempty and identical across **all bidders**; they may vary with $\boldsymbol{\theta}$.
- Every bidder pays her bid cost, and all ties are resolved fairly, including at zero.

Application: All-Pay Auction

Existence, Representation, and Baseline Payoffs

Proposition 1 (Existence and representation in all-pay auctions)

Under Assumption 2, the all-pay auction satisfies Assumption 1. Consequently, it has a BNE, and its equilibria admit the path representation in Theorem 1. In particular, every equilibrium payoff vector satisfies $\boldsymbol{u}^* \geq \mathbf{0}$ *and* $\min_k u_k^* = 0$.

Application: Generalized First-Price Auction

Assumption 3 (Pay-as-bid Auction Primitives)

There are n participating bidders with unit demand and $1 \leq \ell \leq n-1$ homogeneous units for sale.

- Types are finite, with a common prior **allowing arbitrary correlation**. Each bidder knows her private value $0 < v_i(\theta_i) \leq \overline{v} < \infty$.
- Each bidder submits a bid $x_i \geq 0$. The ℓ highest bidders each receive one unit; ties are resolved by fair random ordering, including at zero.
- Winners pay their bid; losers pay a fraction $\varepsilon \in (0, 1]$:

$$u_i^{\varepsilon}(x, r, \boldsymbol{\theta}) = \begin{cases} u_i(v_i(\theta_i) - x), & r \leq \ell, \\ u_i(-\varepsilon x), & r > \ell. \end{cases} \tag{13}$$

- Monetary utility $u_i : \mathbb{R} \to \mathbb{R}$ is continuous and strictly increasing, with $u_i(0) = 0$.

Application: Generalized First-Price Auction

Existence and Equilibrium Payoffs

Proposition 2 (Multi-unit pay-as-bid auctions with private values)

Under Assumption 3, for every $\varepsilon \in (0, 1]$*, the auction has a BNE.*

Application: Costly Bidding

Assumption 4 (Costly-bidding Auction Primitives)

There are n potential bidders with unit demand and $1 \leq \ell \leq n-1$ homogeneous units for sale.

- Types are finite, with a common prior allowing arbitrary correlation. Each bidder knows her private value $0 \leq v_i(\theta_i) \leq \overline{v} < \infty$.
- **Each bidder simultaneously opts out (utility 0, no allocation) or submits $x \geq 0$, paying a fixed fee $\varepsilon > 0$, including at $x = 0$.** Entry decisions are not observed before bidding.
- The highest ℓ submitted bids receive one unit each (all entrants win if fewer than ℓ enter); ties are resolved randomly and fairly.
- Monetary utility $u_i : \mathbb{R} \to \mathbb{R}$ is continuous and strictly increasing, with $u_i(0) = 0$.

Application: Costly Bidding

Existence with a Fixed Submission Fee

For an entrant with rank r among submitted bids,

$$u_i^{\varepsilon}(x, r, \boldsymbol{\theta}) = \begin{cases} u_i\Big(v_i(\theta_i) - x - \varepsilon\Big), & r \le \ell, \\ u_i(-\varepsilon), & r > \ell. \end{cases}$$

Proposition 3 (Existence with a fixed submission fee)

Under Assumption 4, for every $\varepsilon > 0$, the voluntary-entry auction has a BNE.

Contribution of the Existence Results

All-pay Contests with Multiple Ranked Prizes

- Haimanko (2024) establishes existence for single-object all-pay auctions. Proposition 1 accommodates **multiple ranked prizes**, under Assumption 2.
- Our result also gives an **IC-path representation**.

Contribution of the Existence Results

Winner-pay-as-bid with Some Loser-pay

- **ORS: adjust tie breaking.** Olszewski, Reny, and Siegel (2026) study ordinary single-object first-price auctions. Existence is guaranteed by efficient, value-dependent tie breaking (Theorem 1), or by a suitable value-independent rule under regularity (Theorem 4).
- **Our result: existence is guaranteed if there is a small loser-pay component or a small bidding cost.**
- The result allows arbitrary correlation of finite types, continuous strictly increasing monetary utility, and **multiple homogeneous units with unit demand** ($1 \leq \ell < n$).
- This complements ORS through a perturbation in payments; their theorems concern the unperturbed auction ($\varepsilon = 0$).

Beyond the Value-Independent Tie-Breaking Remedy

- For finite values, ORS's regularity condition requires every profile in the product of the marginal supports to have positive probability (Olszewski et al., 2026, p. 199).
- Our result also covers **"nonregular" correlated values**. Consider a two-bidder example where each bidder observes only her own value, and the common prior is uniform on $\{(1, 1), (1, 2), (2, 1)\}$. ▸ Example and proof

Payment modification	Tie-breaking rule	BNE
None	Any rule using only bids and identities	None
εx (losers), $\varepsilon \in (0, 1]$	Fair ties	Exists
Fixed bidding fee $\varepsilon > 0$	Fair ties	Exists

- Thus, a positive loser payment or fixed bidding fee can restore existence even when **no value-independent change in tie breaking can do so**.

Contribution of the Existence Results

- More broadly, our path-based approach provides a unified framework to analyze the equilibrium for such rank-order games.

Extension to General Type Spaces

Both Main Results Extend

Under the counterparts of our assumptions for general type spaces, the **representation result** and the **existence result** continue to hold.

- **Representation.** Every equilibrium can be represented by an incentive-compatible probability path, and every such path generates an equilibrium.
- **Existence.** An incentive-compatible probability path exists, so an equilibrium exists.

The extension covers general measurable type spaces, including discrete, atomless, and mixed type distributions. Correlation is allowed under the information assumption.

Assumptions and results

All-pay Contests with State-dependent Participation

Example: Parameters

- A common-value all-pay contest with two equally likely states: $\rho_h = \rho_\ell = 0.5$.
- The auctioneer randomly invites n_ω bidders from a pool of $N \geq 3$.

	State h	State ℓ
Common value v_ω	1	0.8
Number of participants n_ω	3	2
Good signal $\pi(g \mid \omega)$	0.6	0.4
Bad signal $\pi(b \mid \omega)$	0.4	0.6

- Private signals $\theta \in \{g, b\}$ are independent conditional on the state. Each bidder observes his own signal and invitation, but neither the state nor the number of competitors.

All-pay Contests with State-dependent Participation

Example: Parameters

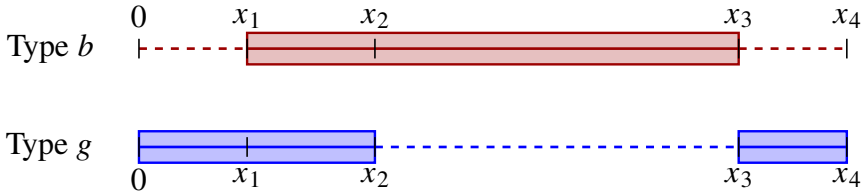
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Good signal $\pi(g \mid \omega)$	0.6	0.4
Bad signal $\pi(b \mid \omega)$	0.4	0.6

- Equilibrium characterization can be very complicated for such state-dependent participation games with common value because a good signal about value carries may also carry bad news regarding competition.

All-pay Contests with State-dependent Participation

Example: Equilibrium Supports

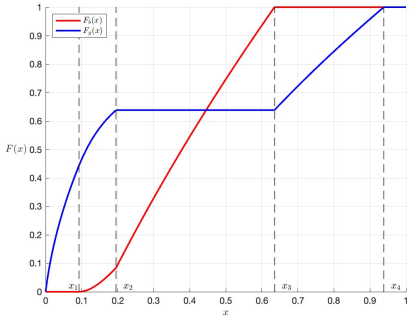


- Equilibrium can be complex: type-g bids have **disconnected support**, $[0, x_2] \cup [x_3, x_4]$. It is hard to guess the support structure, including where each support starts and ends and where the supports overlap.
- Notably, it is **the high type** that **gets a 0 payoff** in equilibrium; the low type's expected payoff is positive.

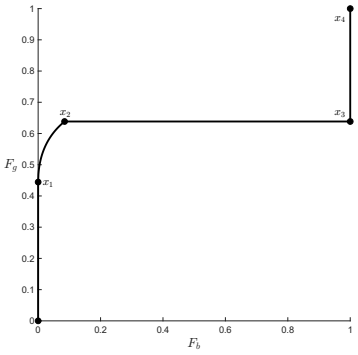
All-pay Contests with State-dependent Participation

Example: Equilibrium

Equilibrium bid distributions



Probability path $(F_b(x), F_g(x))$



- Nonetheless, our approach gives a **full equilibrium characterization** with binary states and signals.

Pure Winner-pay-as-bid

- Jumps in the path (i.e., the two contiguous points on the path corresponding to non-contiguous actions).
  - In the strictly costly rank-order model, the normalized probability path and recovered action are continuous. Every BNE admits such a representation.
  - However, in a more general domain of games, the path need not always increase continuously along at least one coordinate.

  $\Rightarrow$ We need to incorporate "jumps" in the path. We modify the path definition to keep track of $(\boldsymbol{F}^{-}, \boldsymbol{F})$.
- For this path definition, local incentive compatibility must be accompanied by a check of payoffs at ties. The IC paths correspond to BNE of a communication extension of the original game.
  - We further lay out the conditions that determine whether an IC path correspond to a BNE of the original game.

Uniqueness in Complete Information All-pay Contests

Existing Results in the Literature (1/2)

Paper	**Prize structure**	**Cost structure**	**Uniqueness**
Baye et al. (ET, 1996)	Single prize	Heterogeneous linear	Conditional; multiplicity possible
Barut & Kovenock (EJPE, 1998)	Arbitrary prizes	Symmetric linear	Conditional; multiplicity possible
Clark & Riis (AER, 1998)	Homogeneous $v_k = v_{k+1}$	Heterogeneous linear	Unique equilibrium
Bulow & Levin (AER, 2006)	Arithmetic $v_k - v_{k+1} = \beta$	Heterogeneous linear	Unique equilibrium
Siegel (AER, 2010)	Homogeneous $v_k = v_{k+1}$	General, possibly nonlinear costs	Unique equilibrium
González-Díaz & Siegel (IJGT, 2013)	Arithmetic $v_k - v_{k+1} = \beta$	Proportional nonlinear $C_i(s) = \gamma_i C(s)$	Unique via change of variable

Uniqueness in Complete Information All-pay Contests

Existing Results in the Literature (2/2)

Paper	Prize structure	Cost structure	Uniqueness
Xiao (JET, 2016)	Quadratic ($\beta \geq 0$) $v_k - 2v_{k+1} + v_{k+2} = \beta$	Strictly ordered marginal costs	Unique equilibrium (Theorem 1)
	Geometric ($\alpha > 1$) $v_k = \alpha v_{k+1}$ $m = n$, $v_n > 0$	Strictly ordered marginal costs	Unique equilibrium (Theorem 1)

Xiao: differentiable C_i, $C_i(0) = 0$, and $0 < C_1'(s) < \cdots < C_n'(s)$ for every $s \geq 0$.
Quadratic: $k = 1, \ldots, m-1$, $v_{m+1} = 0$; m counts positive prizes. Geometric: $k = 1, \ldots, n-1$.

- Using our approach, we can show that a unique Nash equilibrium exists as long as the prize structure is weakly convex with strictly ordered marginal costs.
- We also discuss the case where marginal costs are weakly ordered.

Summary

- BNE in our specified rank-order games admit a **probability-path representation**.
  - The probability-path together with proposed payoffs contain enough information to check for equilibrium: Fix a path and $\boldsymbol{u} \Longrightarrow$ you can check for IC.
- On the other hand, one can use IC to direct the progress of the path, which boils down to a **shooting problem**, where candidate payoffs determine the implied action and which player–type coordinates can accumulate probability.
- We leverage the shooting perspective to establish the existence of an IC path. Under our assumptions, the representation theorem then guarantees that this path induces a BNE.
- We further fully characterize all equilibria in all-pay contests with state-dependent participation, assuming binary state and binary signals.

Thank You!

References I

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Why Discrete Types?

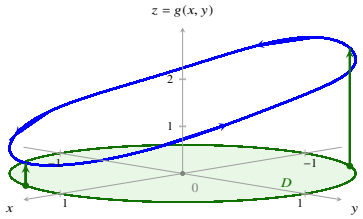
- Why discrete types?
  - Instead of tracing a continuum of types, we need to track finitely many bid distributions.
  - Widely adopted in the literature on auctions and Bayesian games (Harris and Raviv, 1981; Cremer and McLean, 1988; Armstrong, 2000; Akbarpour and Li, 2020; Pernoud and Gleyze, 2023; Murto and Välimäki, 2025).
  - Nest complete-information settings as a special case: each player can be viewed as a "type" that occurs with probability one.
- However, this does ***not*** render the problem easy, as technical obstacles are essentially due to informational and allocative complexities.
  - Supports may still be disconnected, non-monotone, or contain atoms.

Back

Equilibrium Existence

Boundary Dependence of the Brouwer Degree

Let $D \subset \mathbb{R}^n$, $n \geq 1$, be compact with $D = \overline{\operatorname{int} D}$, and let $f : D \to \mathbb{R}^n$ be continuous, with $0 \notin f(\partial D)$.



Scalar illustration in two input dimensions: $g : D \to \mathbb{R}$. Blue arrows orient the graph above ∂D.

Boundary dependence.

$$\boxed{\operatorname{wind}(f(\partial D), 0) = \deg(f, \operatorname{int} D, 0)}.$$

The degree is determined entirely by the boundary map $f|_{\partial D}$. For a C^1 map with regular zeros, it counts solutions *with signs*.

For a planar disk, winding is the *net* number of counterclockwise turns of the image around 0 as ∂D is traversed counterclockwise. In general, wind denotes generalized winding.

Boundary dependence and generalized winding: Vandervorst (2014, Prop. 1.6.1 and Def. 1.6.3, p. 42).

Degree properties used below: Górniewicz (2006, Thm. 26.2 and Prop. 26.6, pp. 124–128).

Equilibrium Existence: Constructing the Degree Map

1. Choose Candidate Payoffs and Measure Which Constraints Bind

Choose a payoff cap above every possible zero-action gain:

$$B_0 := \max_{k,\, \boldsymbol{F} \in [0,1]^K} \left(\widehat{U}_k(0; \boldsymbol{F}) - \beta_k \right), \qquad M > B_0, \qquad \eta := M - B_0 > 0.$$

Search over the $K - 1$ dimensional set

$$\mathcal{U} := \{ \boldsymbol{u} : \boldsymbol{\beta} \le \boldsymbol{u} \le \boldsymbol{\beta} + M\mathbf{1},\ \min_k (u_k - \beta_k) = 0 \}.$$

For trial paths outside the probability cube, let ρ clip each coordinate to $[0, 1]$. Define the recovered action and payoff residuals by

$$L(\boldsymbol{F}, \boldsymbol{u}) := \underline{x}(\rho(\boldsymbol{F}); \boldsymbol{u}),$$
$$a_k(\boldsymbol{F}, \boldsymbol{u}) := \widehat{U}_k \left(L(\boldsymbol{F}, \boldsymbol{u}); \rho(\boldsymbol{F}) \right) - u_k.$$

Then $a_k \le 0$ and $\max_k a_k = 0$: **at least one agent is indifferent**. The exact shooting rule allows only indifferent agents to move.

Equilibrium Existence: Constructing the Degree Map

2. Smooth the Shooting Rule to Obtain a Continuous Endpoint Map

On $C \times \mathcal{U}$, where $C = [0, K]^K$, approximate the continuous residuals by polynomials p_k^m with $\sup |p_k^m - a_k| \le \alpha_m \to 0$. For $\tau_m \downarrow 0$, set

$$G_k^m(\boldsymbol{F}, \boldsymbol{u}) := K \frac{\exp(p_k^m(\boldsymbol{F}, \boldsymbol{u})/\tau_m)}{\sum_j \exp(p_j^m(\boldsymbol{F}, \boldsymbol{u})/\tau_m)}, \qquad \dot{\boldsymbol{F}}^m = G^m(\boldsymbol{F}^m, \boldsymbol{u}), \quad \boldsymbol{F}^m(0) = \mathbf{0}.$$

- Smooth velocities give a unique shot, continuous in $\boldsymbol{u}$. Polynomial approximation ensures this even when the original residuals are only continuous.
- $0 < G_k^m \le K$ and $\sum_k G_k^m = K$, so $\sum_k F_k^m(t) = Kt$ and every shot stays in C.

Thus the endpoint map is continuous:

$$\boxed{v^m : \mathcal{U} \longrightarrow \mathcal{V}, \qquad v^m(\boldsymbol{u}) := \boldsymbol{F}^m(1; \boldsymbol{u})}, \qquad \mathcal{V} := \{v \in \mathbb{R}_+^K : \textstyle\sum_k v_k = K\}.$$

The target is $v^m(\boldsymbol{u}) = \mathbf{1}$.

Equilibrium Existence: Constructing the Degree Map

3. A Boundary Payoff Forces Its Endpoint Coordinate Below One

If $u_k = \beta_k + M$, then agent k's payoff target is uniformly too high:

$$a_k(\boldsymbol{F}, \boldsymbol{u}) \le \widehat{U}_k(0; \rho(\boldsymbol{F})) - \beta_k - M \le B_0 - M = -\eta.$$

Another residual equals zero. Since $|p_j^m - a_j| \le \alpha_m$ on $C \times \mathcal{U}$,

$$G_k^m(\boldsymbol{F}, \boldsymbol{u}) \le K \exp\left(\frac{-\eta + 2\alpha_m}{\tau_m}\right) \le K \exp\left(-\frac{\eta}{2\tau_m}\right) < 1,$$

where we choose $\alpha_m \le \eta/4$ and τ_m small enough for the last inequality. Integrating over $t \in [0, 1]$ gives

$$\boxed{u_k = \beta_k + M \quad \Longrightarrow \quad v_k^m(\boldsymbol{u}) < 1.}$$

This is the boundary information needed for degree: an agent assigned the largest candidate payoff cannot accumulate unit mass.

Equilibrium Existence: Constructing the Degree Map

4. Put Payoff Guesses and Endpoints on the Same Ball

Both $\mathcal{U}$ and $\mathcal{V}$ are $K-1$ dimensional. Use the ball

$$H := \{z \in \mathbb{R}^K : \textstyle\sum_k z_k = 0\} \simeq \mathbb{R}^{K-1}, \qquad D := \{z \in H : \|z\|_2 \le 1\}.$$

Write $\boldsymbol{d} = \boldsymbol{u} - \boldsymbol{\beta}$ and $\bar{d} = K^{-1}\sum_k d_k$. The following radial coordinate changes are homeomorphisms onto D:

$$h_0 : \mathcal{U} \to D, \qquad h_0(\boldsymbol{u}) = \frac{\boldsymbol{d} - \bar{d}\mathbf{1}}{\|\boldsymbol{d} - \bar{d}\mathbf{1}\|_2}\,\frac{\max_k d_k}{M}, \qquad h_0(\boldsymbol{\beta}) = 0,$$

$$h_1 : \mathcal{V} \to D, \qquad h_1(v) = -\frac{v - \mathbf{1}}{\|v - \mathbf{1}\|_2}(1 - \min_k v_k), \qquad h_1(\mathbf{1}) = 0.$$

- h_0 sends $\max_k(u_k - \beta_k) = M$ to ∂D and the baseline $\boldsymbol{\beta}$ to the center.
- h_1 sends the target $\mathbf{1}$ to zero. Its minus sign makes an endpoint shortfall a positive coordinate.

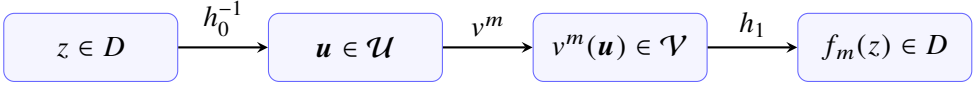
All boundaries and interiors of D below are relative to H.

Equilibrium Existence: Constructing the Degree Map

5. Compose the Coordinate Changes with the Endpoint Map

For each approximation m, the map used in the degree argument is

$$f_m := h_1 \circ v^m \circ h_0^{-1} : D \longrightarrow D.$$



Continuity of all three maps makes f_m continuous. Since $h_1^{-1}(0) = \mathbf{1}$,

$$f_m(z) = 0 \quad \Longleftrightarrow \quad \boldsymbol{F}^m\left(1; h_0^{-1}(z)\right) = \mathbf{1}.$$

A zero of f_m selects payoffs whose shot reaches the desired endpoint.

Equilibrium Existence: Constructing the Degree Map

6. The Boundary Signs Give a Homotopy to the Identity

Take $z \in \partial D$ and $\boldsymbol{u} = h_0^{-1}(z)$. Choose k with $u_k - \beta_k = M$. The *same coordinate* has two positive signs:

$$z_k > 0 \quad \text{because } d_k = M > \bar{d}, \qquad f_m(z)_k > 0 \quad \text{because } v_k^m(\boldsymbol{u}) < 1.$$

Consequently, the straight-line homotopy avoids zero on the boundary:

$$H_s(z) := (1-s)z + s f_m(z), \qquad H_s(z)_k > 0 \quad (z \in \partial D,\ 0 \le s \le 1).$$

Homotopy invariance and normalization now imply

$$\boxed{\deg(f_m, \operatorname{int} D, 0) = \deg(\mathrm{id}, \operatorname{int} D, 0) = 1.}$$

Thus some $z_m \in \operatorname{int} D$ satisfies $f_m(z_m) = 0$. Setting $\boldsymbol{u}^m = h_0^{-1}(z_m)$ gives

$$\boldsymbol{F}^m(1; \boldsymbol{u}^m) = \mathbf{1}.$$

Only the boundary signs were needed; the interior map may be complicated.

Equilibrium Existence: Constructing the Degree Map

7. Pass from Exact Endpoints to an Exact IC Path

Write $\boldsymbol{Q}^m(t) = \boldsymbol{F}^m(t; \boldsymbol{u}^m)$ for the shots selected by degree.

- They are nondecreasing, K-Lipschitz, and join $\mathbf{0}$ to $\mathbf{1}$. Hence they stay in $[0,1]^K$, where clipping is inactive.
- Compactness gives a subsequence with
$$\boldsymbol{u}^m \to \boldsymbol{u} \in \mathcal{U}, \qquad \boldsymbol{Q}^m \to \boldsymbol{Q} \quad \text{uniformly on } [0,1].$$
- If $a_k(\boldsymbol{Q}(t), \boldsymbol{u}) < 0$, then locally, for some $\gamma > 0$ and all large m,
$$0 \le \dot{Q}_k^m \le K \exp\left(-\frac{\gamma}{2\tau_m}\right) \longrightarrow 0.$$
Thus Q_k is constant there: only binding agents move in the limit.

With $\nu_k = \mathrm{d}Q_k$, the limiting path satisfies
$$\widehat{U}_k\Big(\underline{x}(\boldsymbol{Q}(t); \boldsymbol{u}); \boldsymbol{Q}(t)\Big) = u_k \quad (t \in \operatorname{supp} \nu_k), \qquad \boldsymbol{Q}(0) = \mathbf{0}, \quad \boldsymbol{Q}(1) = \mathbf{1}.$$
Also $\sum_k Q_k(t) = Kt$ and $\boldsymbol{u} \in \mathcal{U}$: the limit is an IC path.

Equilibrium Existence: The Illustrative Example

1. Turn Each Payoff Guess into a Continuous Shot

All-pay auction: $v_1 = 1$, $v_2 = 2$. Guess $\boldsymbol{u} \geq \mathbf{0}$ with $\min\{u_1, u_2\} = 0$.

Clip trial coordinates: $c_j = \min\{1, \max\{0, F_j\}\}$. Set

$$L = \max\{0, c_2 - u_1, 2c_1 - u_2\},$$

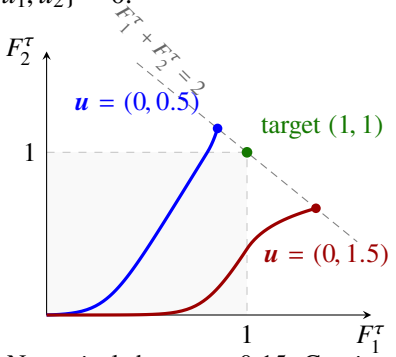
$$a_1 = c_2 - L - u_1, \qquad a_2 = 2c_1 - L - u_2.$$

These are payoff shortfalls: $a_k \leq 0$, $\max_k a_k = 0$.
Replace the binding rule by positive velocities:

$$\dot{F}_k^\tau = 2\frac{e^{a_k/\tau}}{e^{a_1/\tau} + e^{a_2/\tau}}, \qquad \boldsymbol{F}^\tau(0) = \mathbf{0}, \quad \tau > 0.$$

Lipschitz velocities give a unique shot and an endpoint continuous in $\boldsymbol{u}$.

Since $\dot{F}_1^\tau + \dot{F}_2^\tau = 2$, every endpoint lies on the dashed line $F_1^\tau + F_2^\tau = 2$.



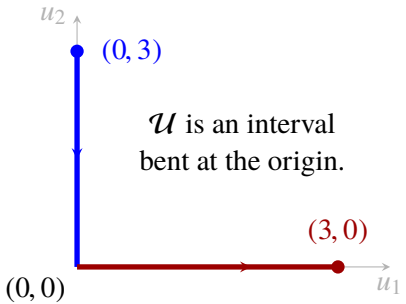
Numerical shots, $\tau = 0.15$. Continue each shot to $t = 1$, including beyond the probability square.

Equilibrium Existence: The Illustrative Example

2. Boundary Payoffs Force the Endpoint to Opposite Sides

The largest zero-bid tie-free payoff is 2. Choose $M = 3$ and search over

$$\mathcal{U} = \{(u_1, u_2) \in [0,3]^2 : \min\{u_1, u_2\} = 0\}.$$



At $u = (0,3)$, player 2's target is too high:

$$a_2 = 2c_1 - L - 3 \le -1, \qquad a_1 = 0.$$

$$\dot{F}_2^\tau < 1 \implies F_2^\tau(1) < 1 < F_1^\tau(1).$$

At $u = (3,0)$, player 1's target is too high:

$$a_1 = c_2 - L - 3 \le -2, \qquad a_2 = 0.$$

$$\dot{F}_1^\tau < 1 \implies F_1^\tau(1) < 1 < F_2^\tau(1).$$

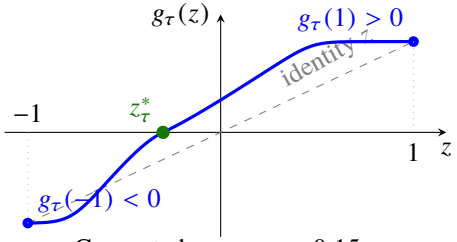
The two ends of the payoff search produce **opposite signs** of the same endpoint error.

Equilibrium Existence: The Illustrative Example

3. Degree 1 Means That the Endpoint Error Must Cross Zero

Here $K - 1 = 1$: degree reduces to the **intermediate value theorem**. For $z \in [-1, 1]$, write $z_+ = \max\{z, 0\}$ and set

$$\boldsymbol{u}(z) = (3z_+, 3(-z)_+), \qquad g_\tau(z) := 1 - F_1^\tau(1; \boldsymbol{u}(z)).$$



Computed curve: $\tau = 0.15$, $\boldsymbol{u}(z_\tau^*) \approx (0, 0.8962)$.

Deform the identity into the continuous error:

$$H_s(z) = (1 - s)z + s\, g_\tau(z), \qquad 0 \le s \le 1,$$

$H_s(-1) < 0 < H_s(1)$: the boundary never hits zero. Hence

$$\deg(g_\tau, (-1, 1), 0) = \deg(\mathrm{id}, (-1, 1), 0) = 1.$$

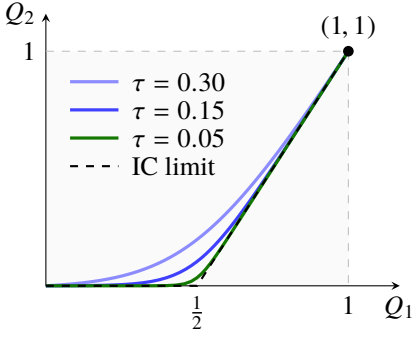
Thus $g_\tau(z_\tau^*) = 0$ for some z_τ^*. The endpoint coordinates sum to 2, so

$$\boldsymbol{F}^\tau(1; \boldsymbol{u}(z_\tau^*)) = (1, 1).$$

Equilibrium Existence: The Illustrative Example

4. Let the Approximation Vanish to Obtain an Exact IC Path

For each τ, select a zero of the endpoint error and write the resulting path as $\boldsymbol{Q}^\tau$.



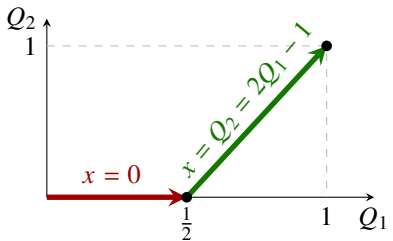
- Exact endpoints and monotonicity keep selected paths inside $[0, 1]^2$. Clipping is inactive.
- As $\tau \downarrow 0$, slopes bounded by 2 give a subsequence with
$$\boldsymbol{u}^\tau \to \boldsymbol{u}^*, \qquad \boldsymbol{Q}^\tau \to \boldsymbol{Q} \quad \text{uniformly.}$$
- If the limiting shortfall is negative near t, then for some $\varepsilon > 0$,
$$0 \le \dot{Q}_k^\tau \le 2e^{-\varepsilon/\tau} \longrightarrow 0.$$
Thus Q_k is constant there: **only indifferent players remain active**.

Equilibrium Existence: The Illustrative Example

5. The Selected Path Recovers the Equilibrium

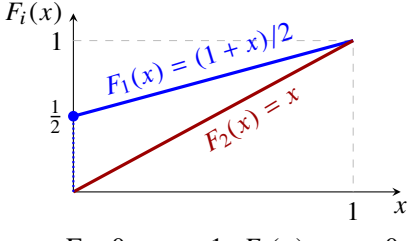
At the end of the shared IC segment, $1 - u_1 = 2 - u_2$. Since $\min\{u_1, u_2\} = 0$, this gives $\boldsymbol{u}^* = (0, 1)$.

Probability path



The first segment gives player 1 an atom of $1/2$ at $x = 0$.

Recovered bid distributions



For $0 < x \leq 1$: $F_2(x) - x = 0$, $2F_1(x) - x = 1$.

At $x = 0$, fair ties give payoffs 0 and $1/2 < 1$; bids $x > 1$ are worse. These strategies therefore form a BNE with payoffs $(0, 1)$.

An Example Beyond Value-Independent Tie Breaking

1. Correlated Private Values

Two risk-neutral bidders, one object, and privately known values. The common prior is

	$v_2 = 1$	$v_2 = 2$
$v_1 = 1$	$1/3$	$1/3$
$v_1 = 2$	$1/3$	0

- A low type (L, value 1) faces each opposing type with probability $1/2$. A high type (H, value 2) always faces a low type.
- Both marginal supports contain 2, but $(2, 2)$ never occurs: **ORS's regularity condition fails**.
- At $\varepsilon = 0$, there is **no BNE under any tie rule based only on bids and bidder identities**, even allowing randomization and asymmetric strategies.

Prior from Olszewski et al. (2026, Remark 3, p. 205), who state nonexistence under fair ties. The next frame establishes the stronger value-independent-tie claim.

An Example Beyond Value-Independent Tie Breaking

2. Why Every Value-Independent Tie Rule Fails at $\varepsilon = 0$

1. **Low types bid at most 1.** Overbids yield nonpositive payoffs. If lows overbid with positive probability, some low wins above value in state $(1, 1)$, whereas bidding zero avoids losses.
2. **High types earn at least 1 and bid at most 1.** A bid $1 + \delta$ wins surely and earns $1 - \delta$. Every bid above 1 gives strictly less than 1.
3. **Both low types must bid 1.** If either bids below 1, the other can profitably outbid some of those bids. Both lows, hence all four types, would then earn positive payoffs. At the smallest support endpoint $m < 1$, positive payoffs require opponent atoms at m, and in turn a winning tie at m between co-occurring types. A type losing that tie can profitably overbid slightly: a contradiction.
4. **Tie priorities conflict.** High types must also bid 1 and win surely. Writing $q = \Pr(1 \text{ wins} \mid x_1 = x_2 = 1)$,

$$(v_1, v_2) = (2, 1) \implies q = 1, \qquad (v_1, v_2) = (1, 2) \implies q = 0.$$

An Example Beyond Value-Independent Tie Breaking

3. An Explicit Equilibrium for Every Positive Loser-Payment Rate

Winners pay x; losers pay εx, where $0 < \varepsilon \leq 1$. Define

$$\lambda := 1 - \varepsilon, \qquad a_\varepsilon := \frac{2}{1 + 2\varepsilon + \sqrt{1 + 8\varepsilon^2}}.$$

Both bidders use the following type-contingent bid CDFs:

$$F_L(x) = \begin{cases} \dfrac{2\varepsilon x}{1 - \lambda x}, & 0 \leq x \leq a_\varepsilon, \\ \dfrac{1 + \varepsilon x}{2 - \lambda x}, & a_\varepsilon \leq x \leq 1, \end{cases} \qquad F_H(x) = \begin{cases} 0, & 0 \leq x < a_\varepsilon, \\ \dfrac{2\varepsilon x}{1 - \lambda x} - \dfrac{1 + \varepsilon x}{2 - \lambda x}, & a_\varepsilon \leq x \leq 1. \end{cases}$$

Both CDFs equal 0 below 0 and 1 above 1.

- These are continuous CDFs. Low types earn 0; high types earn 1.
- There are no ties, even against a unilateral pure-bid deviation. This equilibrium therefore works under **every tie-breaking rule**, including fair ties.

An Example Beyond Value-Independent Tie Breaking

4. Verification and the Limit as Loser Payments Vanish

For $0 \le x \le 1$, a low type's deviation payoff is

$$U_L(x) = (1 - \lambda x)\frac{F_L(x) + F_H(x)}{2} - \varepsilon x = 0.$$

A high type faces a low opponent, so

$$U_H(x) = (2 - \lambda x)F_L(x) - \varepsilon x = \begin{cases} \dfrac{\varepsilon x(3 - \lambda x)}{1 - \lambda x} \le 1, & 0 \le x < a_\varepsilon, \\ 1, & a_\varepsilon \le x \le 1. \end{cases}$$

- Bids above 1 win surely and yield $1 - x < 0$ for a low type or $2 - x < 1$ for a high type. Thus no deviation is profitable.
- As $\varepsilon \downarrow 0$, both bid distributions converge to a point mass at 1. At the limit, the incompatible tie-priority requirements return.

An Example Beyond Value-Independent Tie Breaking

5. An Explicit Equilibrium with a Small Bidding Fee

Keep the same prior. Bidders may opt out for payoff 0; every submitted bid, including zero, costs ε, and the winner additionally pays x. For $0 < \varepsilon \le 1/3$, define

$$a := \frac{1-3\varepsilon}{1-\varepsilon}, \qquad b := 1-\varepsilon.$$

Low types opt out with probability 2ε; high types always enter. Write

$$G_t(x) := \Pr\Big(\text{type } t \text{ opts out or submits a bid } \le x\Big).$$

For $0 \le x \le b$, the strategies are specified by

$$G_L(x) = \begin{cases} \dfrac{2\varepsilon}{1-x}, & 0 \le x \le a, \\ \dfrac{1+\varepsilon}{2-x}, & a \le x \le b, \end{cases} \qquad G_H(x) = \begin{cases} 0, & 0 \le x < a, \\ \dfrac{2\varepsilon}{1-x} - \dfrac{1+\varepsilon}{2-x}, & a \le x \le b. \end{cases}$$

Both functions are nondecreasing and equal 1 at b and above. **Opting out is distinct from submitting a zero bid**; submitted bids have no atoms.

An Example Beyond Value-Independent Tie Breaking

6. Fixed-Fee Verification and the Vanishing-Fee Limit

Against these atomless submitted bids, an entrant's deviation payoffs on $[0, b]$ are

$$U_L(x) = (1 - x)\frac{G_L(x) + G_H(x)}{2} - \varepsilon = 0,$$

$$U_H(x) = (2 - x)G_L(x) - \varepsilon = \begin{cases} \dfrac{\varepsilon(3 - x)}{1 - x} \le 1, & 0 \le x < a, \\ 1, & a \le x \le b. \end{cases}$$

- Low types earn 0 from entering or opting out; high types earn 1 on $[a, b]$ and cannot gain by opting out.
- A bid $x > b$ wins surely and yields $1 - x - \varepsilon < 0$ for a low type or $2 - x - \varepsilon < 1$ for a high type. Thus these strategies form a BNE.
- As $\varepsilon \downarrow 0$, the opting-out probability vanishes and both bid distributions converge to a point mass at 1. The limiting auction again has no BNE under value-independent ties.

An Example Beyond Value-Independent Tie Breaking

7. Larger Fees: Completion of the Existence Proof

- For $1/3 < \varepsilon < 1$, low types opt out with probability $p = (1+\varepsilon)/2$. High types always enter, with an atom $h = (3\varepsilon - 1)/2$ at a submitted bid of zero. Set $b = 1 - \varepsilon$ and
$$G_L(x) = \frac{1+\varepsilon}{2-x}, \qquad G_H(x) = \frac{2\varepsilon}{1-x} - G_L(x), \qquad 0 \le x \le b.$$
These functions are nondecreasing and reach 1 at b. For $0 < x \le b$, the same payoff formulas give $U_L(x) = 0$ and $U_H(x) = 1$. At zero, fair ties give
$$U_L(0) = \frac{p + h/2}{2} - \varepsilon = -\frac{h}{4} < 0, \qquad U_H(0) = 2p - \varepsilon = 1.$$
Bids above b and opting out cannot improve these equilibrium payoffs.
- For $1 \le \varepsilon < 2$, low types opt out and high types submit zero, earning $2 - \varepsilon$. A low entrant earns at most $1 - \varepsilon \le 0$; a high type always faces an opting-out low type, so bidding zero is optimal.
- For $\varepsilon \ge 2$, everyone opts out, since any submitted bid yields at most $2 - \varepsilon \le 0$.

General Type Spaces: Environment

- There are $n \geq 2$ players and $\ell \in \{1, \ldots, n\}$ prizes. All players participate and choose actions $x_i \geq 0$; ties are resolved by the same fair random ordering.
- Player i privately observes s_i in a measurable space $(S_i, \mathcal{S}_i)$. Write $S = \prod_i S_i$ and let P have actual marginals P_i. Rank payoffs are $u_i(x, r, s)$.
- **Assumption 5 (Information):**

$$P \ll P^{\otimes} := \bigotimes_i P_i, \qquad h := \frac{\mathrm{d}P}{\mathrm{d}P^{\otimes}}.$$

  The marginals P_i may have atoms. Correlation is allowed; the density h may vanish or be unbounded.
- A behavioral strategy is a measurable kernel $\sigma_i(\mathrm{d}x \mid s_i)$. BNE requires best responses for P_i-almost every own type, using the player's full information.

General Type Spaces: Assumption 6 (Rank Payoffs)

Rank payoffs are jointly measurable and, for P_i-almost every s_i,

$$\mathbb{E}\left[\max_{1\le r\le n} |u_i(x, r, \boldsymbol{s})| \mid s_i\right] < \infty \quad \text{at each fixed } x \ge 0.$$

For P-almost every $\boldsymbol{s}$, simultaneously for all players, ranks, and actions in the stated ranges:

- At each rank, payoffs are continuous and **strictly decreasing** in x. With a common cutoff $\bar{x} > 0$,

$$u_i(0, r, \boldsymbol{s}) \ge 0, \qquad u_i(x, r, \boldsymbol{s}) < 0 \quad \text{for } x \ge \bar{x}.$$

- On $A := [0, \bar{x}]$, better ranks are weakly preferred:

$$u_i(x, r, \boldsymbol{s}) \ge u_i(x, r+1, \boldsymbol{s}), \qquad r < n.$$

General Type Spaces: Rank Payoffs

Assumption 6 (continued)

For P-almost every type profile $\boldsymbol{s}$, define

$$B_i(x, \boldsymbol{s}) := \Big\{ r \in \{1, \ldots, n-1\} : u_i(x, r, \boldsymbol{s}) > u_i(x, r+1, \boldsymbol{s}) \Big\}.$$

- At each such profile, $B_i(x, \boldsymbol{s})$ is nonempty and the same across all players i and actions $x \in A$.
- The common set may vary with $\boldsymbol{s}$; the sizes of the gains may depend on the player, action, and type profile.
- Every player strictly prefers first to last place, and all members of a tied group agree on which adjacent ranks yield strict gains.

General Type Spaces: Probability Paths

- For each type, let $\eta_i(\mathrm{d}t \mid s_i)$ be a probability kernel on $[0,1]$. The joint type–time measure is
$$M_i(\mathrm{d}s_i, \mathrm{d}t) = P_i(\mathrm{d}s_i)\eta_i(\mathrm{d}t \mid s_i).$$
- Write λ_i for its unconditional time marginal and impose the **probability clock**:
$$\frac{1}{n}\sum_i \lambda_i = \mathrm{Leb}_{[0,1]}.$$
- The accumulated probabilities
$$Q_i(t)(s_i) := \eta_i([0,t] \mid s_i)$$
form a continuous, nondecreasing path in $\prod_i L^1(P_i)$, with $\boldsymbol{Q}(0) = \mathbf{0}$ and $\boldsymbol{Q}(1) = \mathbf{1}$.
- The clock replaces the finite sum of agent coordinates. Individual conditional time distributions may still have atoms.

General Type Spaces: Actions and Path Payoffs

- Let $\phi : [0,1] \to A$ be a measurable, weakly increasing action map. Its **completed graph** is

$$G_\phi := \{(x,t) : \phi(t-) \le x \le \phi(t+)\}, \qquad \phi(0-) = 0, \quad \phi(1+) = \bar{x}.$$

  This includes actions skipped by jumps of ϕ.

- Later positions receive better ranks. If player i chooses position t, her virtual rank is

$$r_i(t, \boldsymbol{t}_{-i}) := 1 + \#\{j \neq i : t_j > t\}.$$

- Her interim payoff from action x and position t is

$$\Pi_i(s_i; x, t) := \int h(\boldsymbol{s})\, u_i\Big(x, r_i(t, \boldsymbol{t}_{-i}), \boldsymbol{s}\Big) \prod_{j \neq i} M_j(\mathrm{d}s_j, \mathrm{d}t_j).$$

  The density h retains the correlation between players' types.

General Type Spaces: Incentive-Compatible Paths

The worst-rank baseline is

$$\beta_i(s_i) := \mathbb{E}[u_i(0, n, \boldsymbol{s}) \mid s_i] \geq 0.$$

Definition (IC probability path). A triple $(\boldsymbol{M}, \phi, \boldsymbol{\nu})$ is incentive compatible if:

- The probability clock holds, and $\nu_i \geq \beta_i$ is finite and measurable almost everywhere.
- On one P_i-full set of own types, **every graph deviation is bounded**:

$$\Pi_i(s_i; x, t) \leq \nu_i(s_i) \qquad \text{for every } (x, t) \in G_\phi.$$

- **Used positions are indifferent**:

$$\Pi_i(s_i; \phi(t), t) = \nu_i(s_i) \qquad \text{for } M_i\text{-almost every } (s_i, t).$$

Its projected strategy is $\sigma_i(\cdot \mid s_i) = \phi_\# \eta_i(\cdot \mid s_i)$: draw a position t and choose action $\phi(t)$.

General Type Spaces: Equilibrium Representation

Theorem 3 (Probability-path representation)

Under Assumptions 5 and 6:

(i) *Every behavioral BNE with interim payoffs* $\boldsymbol{v}$ *admits an IC probability path that projects to its strategies and has the same payoffs.*

(ii) *Every IC probability path* $(\boldsymbol{M}, \phi, \boldsymbol{v})$ *projects to a behavioral BNE under fair ties, with interim payoffs* $\boldsymbol{v}$.

Strategies and payoffs are identified up to own-type null sets. The equilibrium inequalities control all deviations using the original information.

General Type Spaces: Path Existence

Theorem 4 (Existence of an IC probability path)

Under Assumptions 5 and 6, there exists an IC probability path $(\boldsymbol{M}, \phi, \boldsymbol{v})$ *with finite measurable payoffs satisfying* $v_i \geq \beta_i$ *almost everywhere.*

By Theorem 3, its projected strategies form a **behavioral BNE under fair ties**, with the same interim payoffs. Equilibrium actions are strictly below $\bar{x}$ almost surely.

- The construction obtains an IC probability path from finite-type approximations and lifts it to the original information.
- These results require conditional integrability, but no ex ante payoff integrability or continuity in the type variables.